

Lecture 15, Mar 02-2026.

Rationalizing Substitutions.

Example. Evaluate $\int \frac{\sqrt{x+4}}{x} dx$.

Sln. Let $u = \sqrt{x+4}$ then $x = u^2 - 4$. then $du = \frac{1}{2\sqrt{x+4}} dx = \frac{1}{2u} dx$
then $dx = 2u du$.

$$I = \int \frac{\sqrt{x+4}}{x} dx = \int \frac{u(2u) du}{u^2 - 4} = 2 \int \frac{u^2}{u^2 - 4} du.$$

$$\begin{array}{r} u^2 \quad | \quad u^2 - 4 \\ -u^2 + 4 \quad | \quad 1 \\ \hline 4 \end{array} \quad \text{then}$$

$$I = 2 \int \frac{u^2}{u^2 - 4} du = 2 \int \left(1 + \frac{4}{u^2 - 4} \right) du.$$

$$= 2u + 8 \int \frac{1}{u^2 - 4} du.$$

$$= 2u + 8 \int \frac{1}{(u+2)(u-2)} du.$$

$$\frac{1}{(u+2)(u-2)} = \frac{A}{u+2} + \frac{B}{u-2} = \frac{A(u-2) + B(u+2)}{(u+2)(u-2)}$$

$$u=2 \quad \text{then} \quad 1 = 4B \Rightarrow B = \frac{1}{4}.$$

$$u=-2 \quad \text{then} \quad 1 = -4A \Rightarrow A = -\frac{1}{4}$$

then

$$I = 2u + 8 \left\{ -\frac{1}{4} \int \frac{1}{u+2} du + \frac{1}{4} \int \frac{1}{u-2} du \right\}$$

$$I = 2u + 8 \left(-\frac{1}{4} \ln|u+2| + \frac{1}{4} \ln|u-2| \right) + C$$

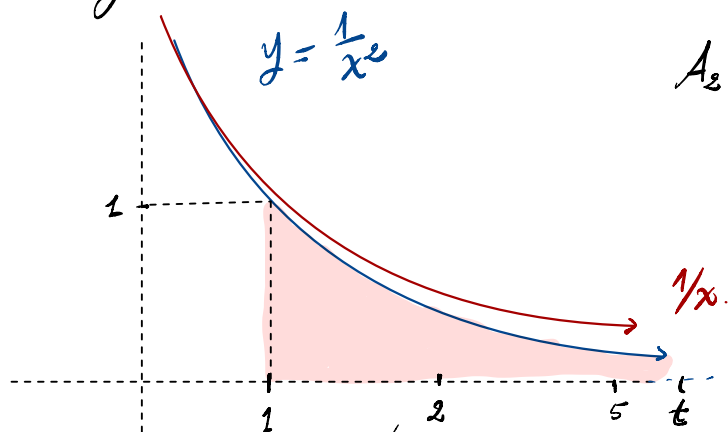
$$I = 2u + 2 \ln \left| \frac{u-2}{u+2} \right| + C$$

$$I = 2\sqrt{x+4} + 2 \ln \left| \frac{\sqrt{x+4}-2}{\sqrt{x+4}+2} \right| + C.$$

7.8. Improper integrals.

Type 1. Infinite intervals.

let consider $y = 1/x^2$, above the x -axis and to the right of $x=1$.
You might think that $\mathcal{B}$ is infinite



$$A_2 = \int_1^2 \frac{1}{x^2} dx = -\frac{1}{x} \Big|_1^2 = \frac{1}{2}.$$

$$A_5 = \int_1^5 \frac{1}{x^2} dx = -\frac{1}{x} \Big|_1^5 = \frac{4}{5}.$$

then $A(t) = \int_1^t \frac{1}{x^2} dx$ this a new function on the t variable.

then, $A(t) \xrightarrow{?} \square$ as $t \rightarrow \infty$. By definition

$$\lim_{t \rightarrow \infty} A(t) = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^2} dx := \int_1^{\infty} \frac{1}{x^2} dx$$

$$\begin{aligned} \text{then } \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^2} dx &= \lim_{t \rightarrow \infty} \left\{ -\frac{1}{x} \Big|_1^t \right\} \\ &= \lim_{t \rightarrow \infty} \left\{ -\frac{1}{t} + 1 \right\} = 1. \end{aligned}$$

$$\begin{aligned} \int_1^{\infty} \frac{1}{x} dx &= \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx \\ &= \lim_{t \rightarrow \infty} \left\{ \ln|x| \Big|_1^t \right\} \\ &= \lim_{t \rightarrow \infty} \ln|t| = \infty \end{aligned}$$

Def. a). If $\int_a^t f(x) dx$ exists for every number $t \geq a$ then.

$$\int_a^{\infty} f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx.$$

b). If $\int_t^b f(x) dx$ exists for every number $t \leq b$, then

$$\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx.$$

The improper integrals $\int_a^{\infty} f(x) dx$, $\int_{-\infty}^b f(x) dx$ are called convergent if the corresponding limit exists and divergent if the limit does not exist.

c). If both $\int_a^{\infty} f(x) dx$ and $\int_{-\infty}^a f(x) dx$ are convergent then

$$\int_{-\infty}^{\infty} f(x) dx = \underbrace{\int_{-\infty}^a f(x) dx}_{\text{case b).}} + \underbrace{\int_a^{\infty} f(x) dx}_{\text{case a).}}$$

Example. Evaluate $\int_{-\infty}^0 x e^x dx$.

Sln: $\int_{-\infty}^0 x e^x dx = \lim_{t \rightarrow -\infty} \int_t^0 x e^x dx$ $u = x \quad dv = e^x$
 $du = 1 \quad v = e^x$

by parts

$$\begin{aligned} &= \lim_{t \rightarrow -\infty} \left\{ x e^x \Big|_t^0 - \int_t^0 e^x dx \right\} \\ &= \lim_{t \rightarrow -\infty} \left\{ -t e^t - e^x \Big|_t^0 \right\} \\ &= \lim_{t \rightarrow -\infty} \left\{ -t e^t - 1 + e^t \right\} \\ &= -1 + \lim_{t \rightarrow -\infty} e^t (1-t) \\ &= -1 + \lim_{u \rightarrow \infty} \frac{1+u}{e^u} \qquad e^{t(1-t)} = \frac{(1-t)}{e^{-t}} \\ &\stackrel{\text{L'H}}{=} -1 + \lim_{u \rightarrow \infty} \frac{1}{e^u} = -1. \end{aligned}$$

Example: Evaluate $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$.

Sln. Under the assumption that $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$ is convergent then

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = \int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^{\infty} \frac{1}{1+x^2} dx.$$

$$\begin{aligned} \bullet \int_{-\infty}^0 \frac{1}{1+x^2} dx &= \lim_{t \rightarrow -\infty} \int_t^0 \frac{1}{1+x^2} dx = \lim_{t \rightarrow -\infty} \left\{ \arctan(x) \Big|_t^0 \right\} \\ &= \lim_{t \rightarrow -\infty} \{ \arctan(0) - \arctan(t) \} \\ &= -\lim_{t \rightarrow -\infty} \arctan(t) = -\frac{\pi}{2} \end{aligned}$$

$$\begin{aligned} \bullet \int_0^{\infty} \frac{1}{1+x^2} dx &= \lim_{t \rightarrow \infty} \left\{ \arctan(x) \Big|_0^t \right\} \\ &= \lim_{t \rightarrow \infty} \arctan(t) = \frac{\pi}{2}. \end{aligned}$$

then $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$ is convergent and

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = \frac{\pi}{2} + \frac{\pi}{2} = \pi.$$

Example: For what values of p is the integral

$$\int_1^{\infty} \frac{1}{x^p} dx$$

convergent?

Sln. $\int_1^{\infty} \frac{1}{x^p} dx = \lim_{t \rightarrow \infty} \int_1^t x^{-p} dx, \quad p \neq 1.$

$$= \lim_{t \rightarrow \infty} \left. \frac{x^{1-p}}{1-p} \right|_1^t$$

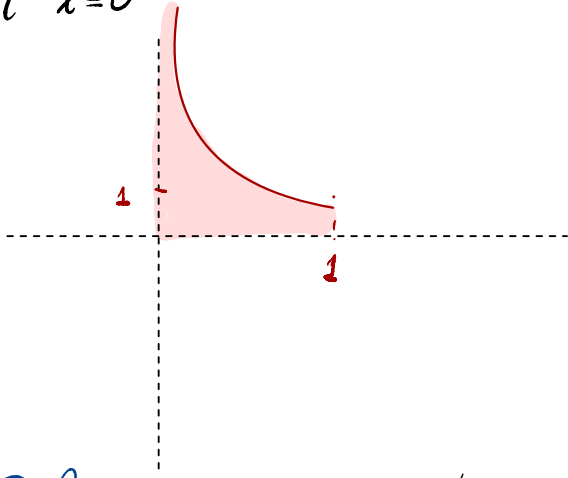
$$= \lim_{t \rightarrow \infty} \left\{ \frac{t^{1-p}}{1-p} - \frac{1}{1-p} \right\}$$

$$= \frac{1}{1-p} \lim_{t \rightarrow \infty} \left\{ \frac{1}{t^{p-1}} - 1 \right\} = \begin{cases} \frac{1}{p-1} & \text{if } p > 1. \\ \text{Divergent} & \text{if } p \leq 1. \end{cases}$$

$\int_2^{\infty} \frac{1}{x^{0.99999}} dx$ this a p -integral with $p=0.99999 < 1$.
 then is divergent.

Type 2. Discontinuous integrands.

For example $\frac{1}{x}$ and $\int_0^1 \frac{1}{x} dx$ and $f(x) = \frac{1}{x}$ is not defined at $x=0$



if x_0 is a point where f is not continuous
 then $\int_a^{x_0} f(x) dx = \lim_{t \rightarrow x_0^-} \int_a^t f(x) dx$.

Def. a). If f is continuous $[a, b)$ and is discontinuous at b , then

$$\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx.$$

if this limit exists.

b). If f is continuous on $(a, b]$ and is discontinuous at a , then

$$\int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx.$$

if this limit exists.

The improper integral $\int_a^b f(x) dx$ is called convergent if the corresponding limit exists and divergent if the limit does not exist.

c). If f has a discontinuity at c , where $a < c < b$ and both $\int_a^c f(x) dx$ and $\int_c^b f(x) dx$ are convergent, then we define

$$\int_a^b f(x) dx = \underbrace{\int_a^c f(x) dx}_{\text{type a).}} + \underbrace{\int_c^b f(x) dx}_{\text{type b).}}$$

Example. Find $\int_2^3 \frac{1}{\sqrt{x-2}} dx$

Sln. $2 \notin \text{Dom}\left(\frac{1}{\sqrt{x-2}}\right)$. then

$$\begin{aligned}\int_2^3 \frac{1}{\sqrt{x-2}} dx &= \lim_{t \rightarrow 2^+} \int_t^3 \frac{1}{\sqrt{x-2}} dx \\&= \lim_{t \rightarrow 2^+} \left. 2\sqrt{x-2} \right|_t^3 \\&= \lim_{t \rightarrow 2^+} \{2\sqrt{3-2} - 2\sqrt{t-2}\} \\&= 2 \lim_{t \rightarrow 2^+} \{1 - \sqrt{t-2}\} = 2.\end{aligned}$$

Example: i). $\int_0^{\pi/2} \sec x dx$
ii). $\int_0^3 \frac{dx}{x-1}$